

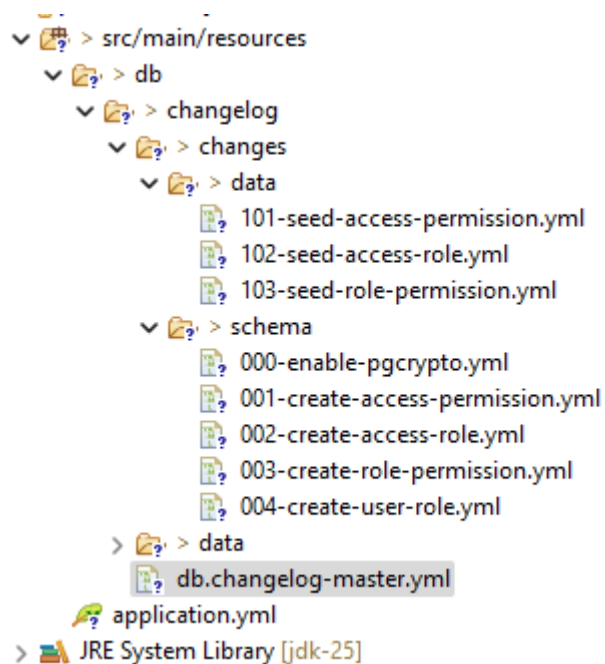
Liquibase Setup

1. Organize the files

Use this structure:

```
src/main/resources/db/
├─ changelog/
│  └─ changes/
│     └─ schema/
│        ├── 000-enable-pgcrypto.yml
│        ├── 001-create-access-permission.yml
│        ├── 002-create-access-role.yml
│        ├── 003-create-role-permission.yml
│        └── 004-create-user-role.yml
│     └─ data/
│        ├── 101-seed-access-permission.yml
│        ├── 102-seed-access-role.yml
│        └── 103-seed-role-permission.yml
│  └─ data/
│     ├── access-permissions.csv
│     ├── access-roles.csv
│     └── role-permissions.csv
└─ db.changelog-master.yml

application.yml
```



The screenshot shows a file explorer view of a project. The structure is as follows:

- src/main/resources
 - db
 - changelog
 - changes
 - data
 - 101-seed-access-permission.yml
 - 102-seed-access-role.yml
 - 103-seed-role-permission.yml
 - schema
 - 000-enable-pgcrypto.yml
 - 001-create-access-permission.yml
 - 002-create-access-role.yml
 - 003-create-role-permission.yml
 - 004-create-user-role.yml
 - data
 - db.changelog-master.yml
 - application.yml

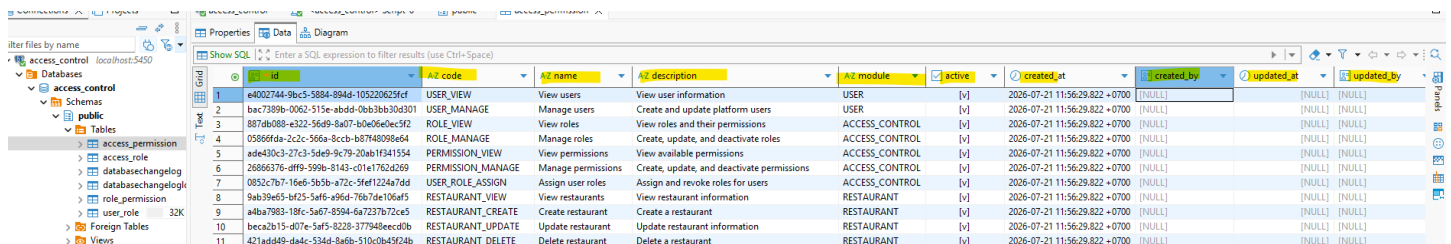
At the bottom of the explorer, there is a link to the JRE System Library [jdk-25].

1-Setup application.yml

```
1 spring:
2   application:
3     name: access-control-service
4   r2dbc:
5     url: r2dbc:postgresql://${DB_HOST:localhost}:${DB_PORT:5450}/${DB_NAME:access_control}
6     username: ${DB_USERNAME:platformuser}
7     password: ${DB_PASSWORD:Platform@2026}
8   liquibase:
9     enabled: true
10    url: jdbc:postgresql://${DB_HOST:localhost}:${DB_PORT:5450}/${DB_NAME:access_control}
11    user: ${DB_USERNAME:platformuser}
12    password: ${DB_PASSWORD:Platform@2026}
13    change-log: classpath:db/changelog/db.changelog-master.yml
14
15 server:
16   port: ${SERVER_PORT:8097}
17
18 logging:
19   level:
20     com.pisethjavaschool.accesscontrol: INFO
21
```

```
liquibase:
  enabled: true
  url: jdbc:postgresql://${DB_HOST:localhost}:${DB_PORT:5450}/${DB_NAME:access_control}
  user: ${DB_USERNAME:platformuser}
  password: ${DB_PASSWORD:Platform@2026}
  change-log: classpath:db/changelog/db.changelog-master.yml
```

2-In the db



ID	Name	Description	Module	Active	Created At	Created By	Updated At	Updated By
e4002744-9bc3-5884-b94d-105220623cf	USER_VIEW	View user information	USER	[v]	2026-07-21 11:56:29.822 +0700	[NULL]	[NULL]	[NULL]
b4c7309b-0062-515e-ab4d-0b33b30d301	USER_MANAGE	Create and update platform users	USER	[v]	2026-07-21 11:56:29.822 +0700	[NULL]	[NULL]	[NULL]
8b74b0b9-a232-56a9-8a07-b0a0e0e4c92	ROLE_VIEW	View roles and their permissions	ACCESS_CONTROL	[v]	2026-07-21 11:56:29.822 +0700	[NULL]	[NULL]	[NULL]
05959fda-2c2c-566a-8ccb-b8748098e64	ROLE_MANAGE	Create, update, and deactivate roles	ACCESS_CONTROL	[v]	2026-07-21 11:56:29.822 +0700	[NULL]	[NULL]	[NULL]
ade430c3-27c3-5da9-9c79-20ab1f341554	PERMISSION_VIEW	View available permissions	ACCESS_CONTROL	[v]	2026-07-21 11:56:29.822 +0700	[NULL]	[NULL]	[NULL]
26866376-dff9-599b-8143-c01e17624269	PERMISSION_MANAGE	Create, update, and deactivate permissions	ACCESS_CONTROL	[v]	2026-07-21 11:56:29.822 +0700	[NULL]	[NULL]	[NULL]
0852c7b7-16e6-5b5b-a72c-5fef1224a7dd	USER_ROLE_ASSIGN	Assign and revoke roles for users	ACCESS_CONTROL	[v]	2026-07-21 11:56:29.822 +0700	[NULL]	[NULL]	[NULL]
9ab39e65-bf25-5af6-a96d-76b7de106af5	RESTAURANT_VIEW	View restaurant information	RESTAURANT	[v]	2026-07-21 11:56:29.822 +0700	[NULL]	[NULL]	[NULL]
a4ba7983-18fc-5af6-8594-6a7237b72ce5	RESTAURANT_CREATE	Create a restaurant	RESTAURANT	[v]	2026-07-21 11:56:29.822 +0700	[NULL]	[NULL]	[NULL]
beca2b15-d07e-5af5-8228-377948eecd0b	RESTAURANT_UPDATE	Update restaurant information	RESTAURANT	[v]	2026-07-21 11:56:29.822 +0700	[NULL]	[NULL]	[NULL]
421add49-da4c-534d-8a6b-510c0b49f24b	RESTAURANT_DELETE	Delete a restaurant	RESTAURANT	[v]	2026-07-21 11:56:29.822 +0700	[NULL]	[NULL]	[NULL]

3-Sample create table permissions

```
databaseChangeLog:
- changeSet:
  id: 001-create-access-permission
  author: ingpiseth
  changes:
    - createTable:
      tableName: access_permission
      columns:
        - column:
            name: id
            type: uuid
            defaultValueComputed: gen_random_uuid()
            constraints:
              primaryKey: true
              nullable: false
        - column:
            name: code
            type: varchar(100)
            constraints:
              nullable: false
        - column:
            name: name
            type: varchar(150)
            constraints:
              nullable: false
        - column:
            name: description
            type: varchar(500)
        - column:
            name: module
            type: varchar(50)
            constraints:
              nullable: false
        - column:
            name: active
            type: boolean
            defaultValueBoolean: true
            constraints:
              nullable: false
        - column:
            name: created_at
            type: timestamp with time zone
            defaultValueComputed: CURRENT_TIMESTAMP
            constraints:
              nullable: false
        - column:
            name: created_by
            type: uuid
        - column:
            name: updated_at
            type: timestamp with time zone
        - column:
            name: updated_by
            type: uuid
    - addUniqueConstraint:
      tableName: access_permission
      columnNames: code
      constraintName: uk_access_permission_code
    - createIndex:
      tableName: access_permission
      indexName: idx_access_permission_module_active
      columns:
        - column:
            name: module
        - column:
            name: active
```

4-Example insert data

```
databaseChangeLog:
- changeSet:
  id: 001-seed-access-permission
  author: ingpiseth
  changes:
    - loadData:
      file: ../../data/access-permissions.csv
      relativeToChangelogFile: true
      tableName: access_permission
      separator: ","
      quotchar: ''
```

5-Db.changelog-master.yml

```
databaseChangeLog:
# Database extensions
- include:
  file: changes/schema/000-enable-pgcrypto.yml
  relativeToChangelogFile: true

# Schema
- include:
  file: changes/schema/001-create-access-permission.yml
  relativeToChangelogFile: true

- include:
  file: changes/schema/002-create-access-role.yml
  relativeToChangelogFile: true

- include:
  file: changes/schema/003-create-role-permission.yml
  relativeToChangelogFile: true

- include:
  file: changes/schema/004-create-user-role.yml
  relativeToChangelogFile: true

# Seed data
- include:
  file: changes/data/101-seed-access-permission.yml
  relativeToChangelogFile: true

- include:
  file: changes/data/102-seed-access-role.yml
  relativeToChangelogFile: true

- include:
  file: changes/data/103-seed-role-permission.yml
  relativeToChangelogFile: true
```